

MISHKA SINGH

Cape Town, Western Cape | 084 892 7164 | mishkasingh117@gmail.com |

A 3rd year Electrical Engineering student at the University of Cape Town. I hope to be able to use the skills and knowledge I acquire at university to give back to this beautiful country I call home. I love to learn new things and challenge my abilities. I have a passion for all things tech and enjoy spending time in nature when I can.

EXPERIENCE

DEC 2024



IEEE GLOBECOM CONFERENCE VOLUNTEER

Assisted in the registration and admin side during the conference. Learned valuable insight into 5G and upcoming communications technologies from industry leaders.

CHPC STUDENT CLUSTER COMPETITION-TEAM CAPTAIN

DEC 2024



NATIONAL ROUND:

-Designed a 3-node cluster- picked CPU, storage, memory, cabling and networking requirements.
-Servers were delivered in person at the start of the round and had to be set up from scratch- connected them together using a switch, installed operating system (Rocky Linux 9.4), created virtual network using ZeroTier to be able to remotely access cluster through Windows PowerShell using our laptops, and set-up all pre-requisite software for benchmarks.
-Various benchmarks were installed and ran across the cluster in real time. This required us to optimise our cluster in the moment to produce the fastest and best results.

-Placed 4th and won the Fan Favourite award.

-First ever all Women of Colour team to participate.

-Pre-round project: Had to create and set up a cluster using Microsoft Azure. Created our own github repository for the set-up instructions, which can be used by future competition participants.

JULY 2024



SELECTION ROUND:

-A week-long competition where, in teams of 4, we learned about High Performance Computing and then had to design a cluster optimised for a specific software (MLPerf). This included setting up the software and designing the hardware (CPU, GPU, Storage specifications etc.) for the cluster.

-Created a PowerPoint of our design and presented the idea to a panel of judges at the end of the week.

-Created virtual cluster (3 nodes) using OpenStack and used Windows PowerShell to access cluster.

-Rocky Linux 9.3: Set up basic workings of cluster (network, system configurations etc.)

NOV 2023 -
DEC 2023



VACATION WORK INTERNSHIP, BVI CONSULTING ENGINEERS

-Updated ongoing projects in AutoCad and Relux software.

-Performed basic onsite checks.

-Edited Word and Excel documents.

EDUCATION

FEB 2023 -
CURRENT



BSC ELECTRICAL AND COMPUTER ENGINEERING, UCT: 3RD YEAR

-Consistently received the Dean's List award for maintaining a high grade point average and academic excellence.

-Can use KiCad, SolidWorks and LTSpice software.

-Code in python and Java.

2022



NATIONAL SENIOR CERTIFICATE, FISH HOEK HIGH SCHOOL

-Metro South District 2022 NSC Awards: 1st in Computer Applications Technology in the district.

-Deputy Head Student Award.

-2nd Place- Best Speaker Competition.

-Honours Blazer (>88% Aggregate).

SKILLS

- Git and Github
- Ability to collaborate effectively in a team
- Excellent communication skills, both verbal and written
- Proficient across Microsoft suite

ACTIVITIES

- Academic Mentor to multiple students in EBE at UCT
- Part of UCT Hiking club